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ABOUT ME

Software Engineer with 5 years of experience in the IT field at various companies, specializing in cloud-native systems and system reliability. Experienced in developing **microservices (Go, JavaScript, Python)** and managing the full application lifecycle using **Docker** and **Kubernetes** across AWS and Azure, implementing **CI/CD pipelines** and enhancing system traceability through observability tools, such as **OpenTelemetry** and **Prometheus**.

OPENSOURCE CONTRIBUTION

ECLIPSE KUKSA DATABROKER | RUST, gRPC

Contributed to the [Eclipse kuksa-databroker](#) by integrating **OpenTelemetry for distributed tracing**. This improved system observability, **enabled trace analysis across microservices**, and helped identify latency bottlenecks in automotive data communication.

EXPERIENCE

ROBERT BOSCH MOBILITY

AUG 2025 – MAR 2026

MASTER THESIS: FAULT INJECTION MECHANISMS IN VIRTUALIZED ENVIRONMENTS AND ITS OBSERVABILITY | Go, gRPC, Docker, Kubernetes, eBPF, Prometheus, Thanos, AWS (EC2, S3 bucket, Lambda), React, TypeScript, Tailwind CSS, Linux operating system, Windows operating system [🔗](#)

- Architected a **platform independent** workflow for **observability framework** designed to monitor the compute node resource utilization (**for Windows, Linux, Docker Containers**) that dictate simulation performance.
- This solution helped in choosing right resources for a given simulation use case, enabling efficient usage of resources and, hence, **reducing operations cost**.
- Deployed cross-platform **observability exporters** (Node, Process, Windows, custom eBPF) as Kubernetes Daemon Sets and configured dynamic discovery and label-based controls to manage resource overhead.
- Developed a **full-stack observability suite** featuring interactive **metrics visualization** dashboard with **log aggregation** and **fault injection**.
- Engineered a fault injection framework to introduce **controlled resource contention** and analyzed its impact on vECU simulations using the observability suite.

ROBERT BOSCH CORPORATE RESEARCH

FEB 2025 – JULY 2025

INTERN | RUST, Go, Docker, OpenTelemetry, Grafana, WebRTC, Microsoft Azure

Low-latency communication strategies to enable Vehicle-to-Cloud (V2C) teleoperation. The goal is to reduce end-to-end latency in teleoperation of vehicles by evaluating and integrating WebRTC-based communication for real-time streaming and signaling.

- Evaluating **Microsoft Azure** infrastructure components including:
 - **Proximity placement groups** to ensure physical co-location of compute resources
 - Evaluating the effect of **VNet** Peering for Low-Latency Interconnects
 - Architectural Considerations for Cloud **Scalability** and **Fault Tolerance**

ROBERT BOSCH CORPORATE RESEARCH

FEB 2024 – NOV 2024

WORK STUDENT | RUST, Go, Docker, OpenTelemetry, Jaeger, Grafana, Prometheus, eBPF, Linux Networking, ROS2, gRPC

- Multi-Platform Framework for Vehicular Functions: Developed a software framework to deploy a single codebase for vehicular functions across **simulation, model, and full-scale ROS based vehicle (FlexCAR)**, significantly **reducing development time by 66.66%**.
- Developed a VSS-to-ROS communication bridge using a **Raspberry Pi to enable non-invasive teleoperation** and **GStreamer video feedback** from a remote-control center.
- Configured the networking around the teleoperation of flexcar inside Arena (intranet) by **allocating dedicated network band**, maintaining the latency to be well within **20ms (end-to-end intranet latency)**, **reduction in latency by 90%**.
- Testing of the kuksa-databroker by modifying a ghz tool (**Go-lang**) to capture custom end-to-end latency metrics.

- Implemented a comprehensive **OpenTelemetry pipeline with Jaeger** to track function spans and trace contexts for unified performance observability.
- Integrated **Prometheus** to collect **service-level metrics**, exposing custom application performance data alongside tracing streams.

ENGINECAL TECHNOLOGY PRIVATE LIMITED, BENGALURU

FEB 2022 – FEB 2023

SOFTWARE ENGINEER | *JavaScript, TypeScript, Python, MEAN stack, Tailwind CSS, AWS, Microsoft Azure, Docker, Kubernetes*

Worked as a Product development Engineer mainly focused on DevOps and Full-stack development.

Developed and maintained a microservices architecture for vehicle data processing with **CI/CD**, deployed on **Azure Kubernetes Service (AKS)** and **AWS Elastic Kubernetes Service (EKS)**.

ELLIPSONIC INDIA SOLUTIONS PVT LTD, BENGALURU

APR 2021 - JAN 2022

SOFTWARE ENGINEER | *TypeScript, JavaScript, React, Node.js, Prisma ORM, PostgreSQL, AWS, MERN*

Developed and maintained full-stack web applications using the MERN stack (MongoDB, Express.js, React, Node.js).

Managed cloud infrastructure with **AWS EC2**, **AWS Lambda**, **AWS CloudFront**.

Designed a web application to control AWS instances via **AWS CLI** with real-time status updates of EC2 instances. Used **AWS CloudWatch** and **AWS Lambda** for event-driven automation on AWS infrastructure.

Contributed to building a virtual platform for the **Ministry of Health, Singapore**. The platform included the functionality of sending email which made use of **Amazon Simple Email Service (SES)**, **Amazon CloudWatch**, **AWS Lambda**.

Performed **data engineering** and **backend tasks** in Microsoft Azure, **PostgreSQL** for database design and queries, and working with **Azure Synapse** Analytics to implement **ETL pipelines** using:

- Azure Data Factory** for orchestrating data workflows,
- Azure Data Lake Storage** for scalable storage of structured and unstructured data,
- Synapse SQL & Spark pools** for data transformation and analysis of large datasets in preparation for analytics workflows.

PROJECTS

Observability-framework (Master Thesis) 	Observability of Fault Injection in Virtualization Environments
Distributed-tracing 	Opentelemetry demonstration using custom grpc clients connected to kuksa-databroker
VSS-ROS Bridge 	Acts as a bridge between COVESA Vehicle Signal Specification and ROS messages.
Signaloid-performance-benchmark (GitHub Actions) 	Automating performance benchmark of signaloid API using GitHub Actions
terraform-ansible-ha-cluster 	Local High-Availability 3-tier web simulation using Terraform and Ansible for zero-downtime rolling updates.
FastAPI-OAuth 	RESTful API built with FastAPI and SQLAlchemy. This project provides a complete foundation for user management and secure authentication

SKILLS

Languages: Python, JavaScript, TypeScript, Go, Rust

Technologies & Tools: FastAPI, Docker, Kubernetes, Terraform, Ansible, AWS, Microsoft Azure, OpenTelemetry, Prometheus, SQL, NoSQL, MERN Stack, MEAN Stack, gRPC, REST, ROS, JSON, Agentic Engineering.

EDUCATION

UNIVERSITY OF STUTTGART

APR 2023 – MAR 2026

M.SC ELECTRICAL ENGINEERING – SMART SYSTEMS

CGPA 8.6

BANGALORE INSTITUTE OF TECHNOLOGY

AUG 2018 – AUG 2021

B.E ELECTRONICS AND COMMUNICATIONS

CGPA 8.98

LANGUAGE PROFICIENCY

English: C2, **German:** B1 (Learning)

